

Design Proposal: Silicon Photonics Mach Zehnder Interferometers

CHARLES-DAVID QUIGLEY (CQUIGLEY@CIENA.COM)

Ciena, 505 Bd du Parc Technologique, Québec, QC G1P 4S9

@ 2026 UBC Phot1x: Silicon Photonic Design, Fabrication & Data Analysis

1. Introduction

The Mach Zehnder Interferometer (MZI) is a device that splits light into two paths, then recombines it after some propagation length. The recombination creates interference between the two electrical fields, which result in an output field amplitude that depends on the phase of the light in each path. This property makes the MZI a versatile device that is widely used as an optical filter, switch, modulator or test structure. In the context of the UBC Phot1x course, we propose to fabricate MZI variants that sweeps the path length difference between MZI arms. From the experimental results, we aim to extract the waveguide group index and characterize fabrications imperfections (propagation loss, coupler imbalance, etc.).

2. Theory

Consider the following MZI interferometer schematic:

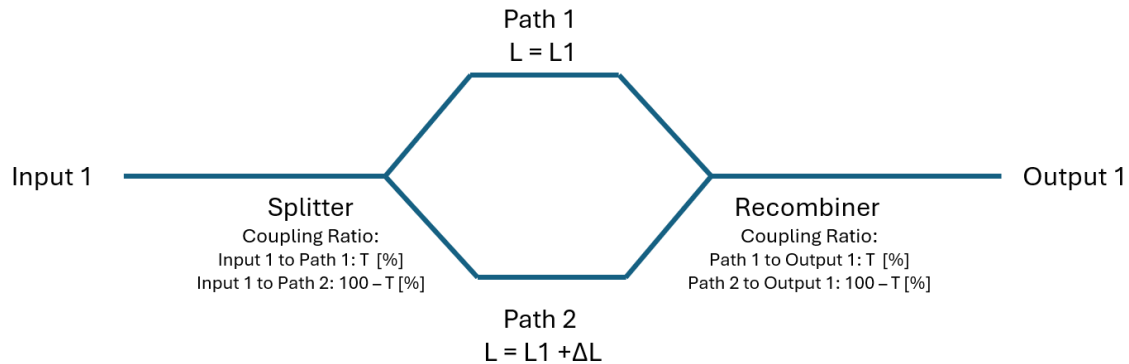


Figure 1: Mach Zehnder Interferometer Schematic

MZI transfer function

If we neglect propagation loss, the input electrical field is:

$$E_{input} = E_0 e^{-i\beta L}$$

The electrical field at the end of each arm is:

$$E_1 = \sqrt{T_{coupling}} E_0 e^{-i\beta_1 L1}$$

$$E_2 = \sqrt{1 - T_{coupling}} E_0 e^{-i\beta_2 (L1 + \Delta L)}$$

Assuming an anti-symmetric coupling ratio for the splitter and combiner ($T_{11_splitter} = T_{21_combiner} = T_{coupling}$ and $T_{12_splitter} = T_{11_combiner} = 1 - T_{coupling}$),

The output 1 electrical field is:

$$E_{out} = \sqrt{1 - T_{coupling}} E_1 + \sqrt{T_{coupling}} E_2 = \sqrt{T_{coupling}(1 - T_{coupling})} E_0 (e^{-i\beta_1 L_1} + e^{-i\beta_2(L_1 + \Delta L)})$$

And the output 1 optical intensity is:

$$I_{out} = |E_{out}|^2 = T_{coupling}(1 - T_{coupling}) I_{input} |e^{-i\beta_1 L_1} + e^{-i\beta_2(L_1 + \Delta L)}|^2$$

Which after using a few trigonometric identities, converts to:

$$I_{out} = 2 T_{coupling}(1 - T_{coupling}) I_{input} (1 + \cos(L_1 \Delta\beta - \beta_2 \Delta L))$$

where

$$\Delta\beta = \beta_1 - \beta_2$$

Note that the complementary output is not considered.

Assuming

- identical waveguides and propagation constants β in both arms
- 50/50 % coupling at the splitter and recombiner

we obtain:

$$I_{out} = 0.5 I_{input} (1 + \cos(\beta \Delta L))$$

MZI Free Spectral Range (FSR)

The MZI FSR is the wavelength or frequency range between two transmission maximums:

$$FSR = \lambda_m - \lambda_{m=1}$$

When propagation constants β are the same in both arms, the transmission will be maximized when

$$\beta \Delta L = m 2\pi$$

$$\beta_m \Delta L - \beta_{m+1} \Delta L = 2\pi$$

$$\Delta\beta = \frac{2\pi}{\Delta L}$$

Using the approximation

$$\Delta\beta \approx -\frac{d\beta}{d\lambda} \Delta\lambda$$

with

$$\frac{d\beta}{d\lambda} = -\frac{2\pi}{\lambda^2} \left(n_{eff} - \frac{dn}{d\lambda} \lambda \right) = -\frac{2\pi}{\lambda^2} (n_g \lambda)$$

We find

$$FSR = \Delta\lambda = \frac{\lambda^2}{n_g \Delta L}$$

3. Proposed layout and simulation

The proposed layout contains

- 5x MZIs of increasing ΔL . They will allow extraction of the waveguide group index.
- 3x test structures to assess coupling loss repeatability and allow coupling loss de-embedding.

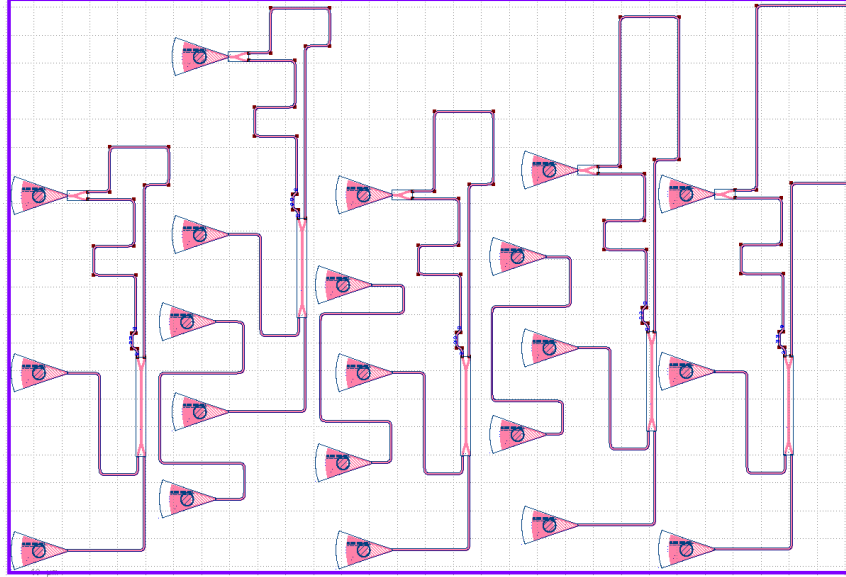


Figure 2 Layout submitted (*EBeam_CharlesDavidQuigley.gds*)

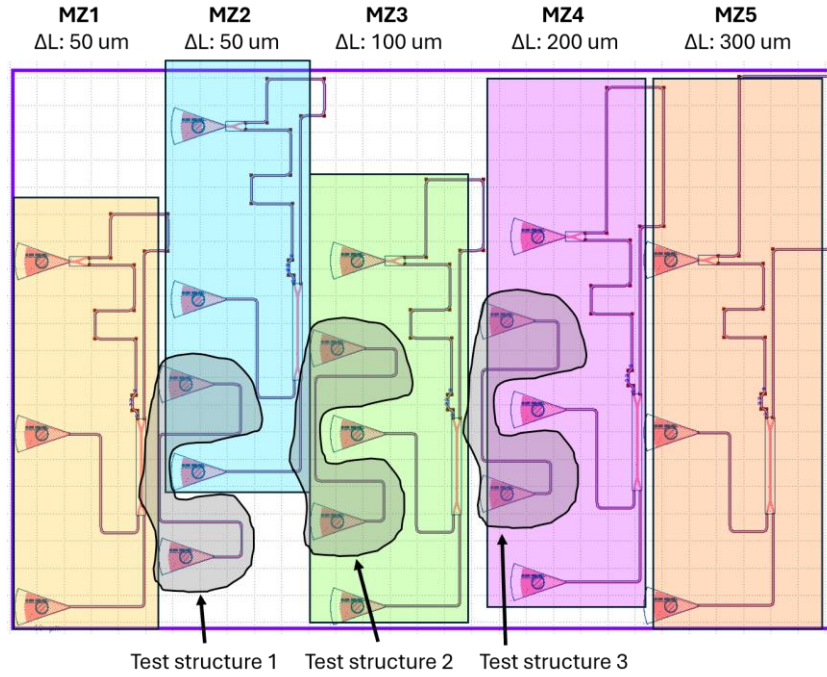


Figure 3 Circuits in layout submitted (*EBeam_CharlesDavidQuigley.gds*)

The Si waveguide dimensions used are height: 220 nm, width: 500 nm, across all structures. The fabrication process and requirements will be described in detail in section 4: fabrication process. Figure 4 shows the cross-section of the fabricated die.

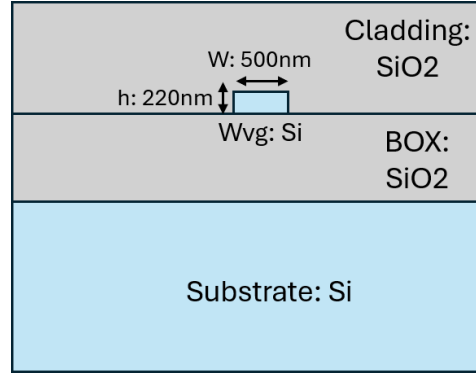


Figure 4 Cross-section of silicon on insulator wafer

We use Lumerical MODE (eigenmode solver) to determine the number of guided modes at 1550 nm for our waveguide configuration. The mode field profile, group index, effective index and other parameters are also extracted for each guided mode using Lumerical MODE. The Si and SiO₂ optical properties used for the simulations are those provided in Handbook of Optical Constants of Solids, Edward D. Palik. The simulation results are summarized in table 1 and figure 5 and 6. Note that all modes for which n_{eff} is below the SiO₂ cladding index ($n \approx 1.444$) are not guided.

Mode ID	n_{eff} @1550 nm	TE (in-plane) polarization fraction [%]	Mode effective area @1550 nm [μm^2]	Guided?
1 (quasi-TE 0)	2.444	98	0.190	Yes
2 (quasi-TM 0)	1.771	4	0.361	Yes
3 (quasi-TE 1)	1.493	70	0.873	Yes
4	1.417	100	7.237	No
5	1.411	15	4.165	No

Table 1: main mode parameters @ 1550nm for 220nm x 500nm strip waveguide

The grating couplers used are optimized for coupling into TE polarized modes. The fact that a 2nd TE mode is guided at 500 nm width should not cause excessive propagation loss, as loss was characterized to be under 3.5dB/cm with this width in previous MPW runs.

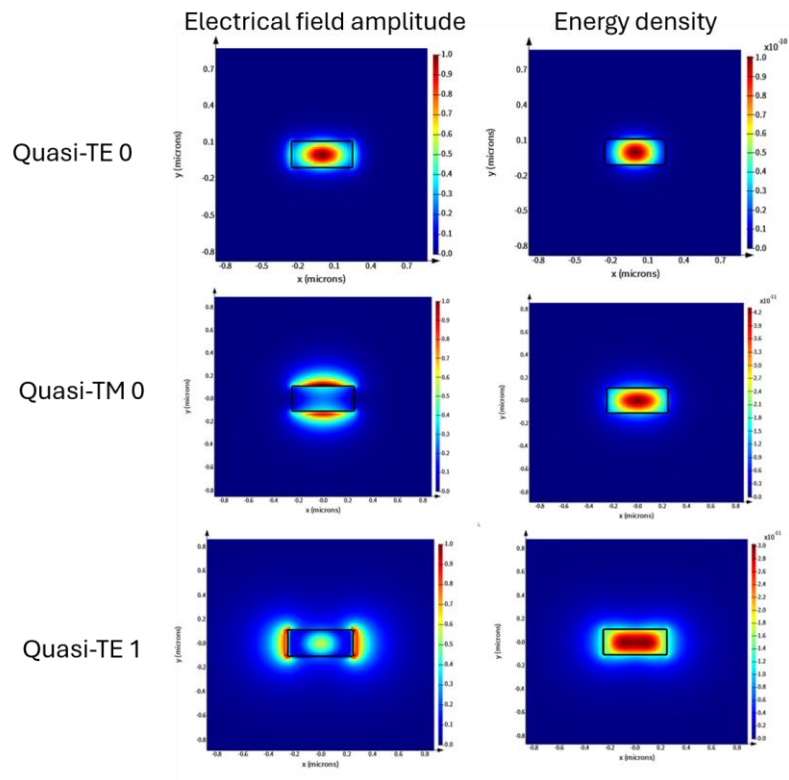


Figure 5 Electrical Field (left) and Energy Density (right) profiles for modes guided in 220nm x 500nm strip waveguide

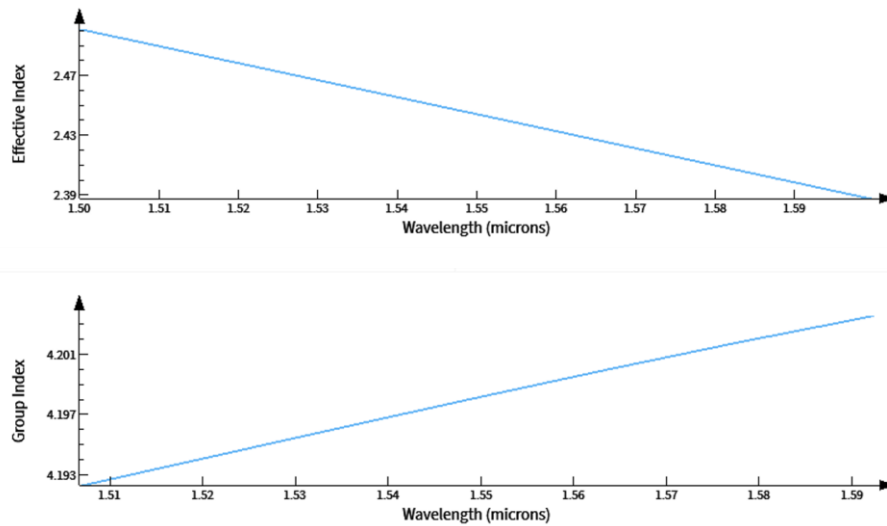


Figure 6 Effective and group index vs wavelength for quasi-TE 0 mode (220nm x 500nm strip waveguide)

Figure 7 shows the expected MZI FSR as a function of MZI arm path length difference at 1550nm, using $n_g = 4.2$.

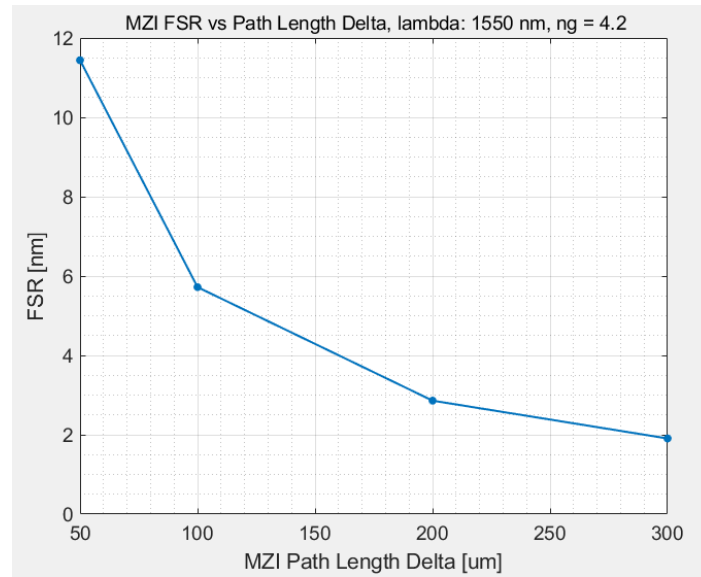


Figure 7 expected MZI FSR as a function of MZI arm path length difference, using $n_g = 4.2$.

4. Fabrication process

To be included in final report.

5. Experimental Results And Analysis

To be included in final report.

6. Summary

To be included in final report.